

Feature Selection in Pre-Diagnosis Heart Coronary Artery Disease Detection

A heuristic approach for feature selection based on Information Gain Ratio and Gini Index

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Abstract— Cardiovascular disease is one of the most common causes of mortality in the world. Among the different types of this disease, the coronary artery is the most important, which the correct and timely diagnosis of which is vital. Diagnostic and treatment methods of this disease have many side effects and costs. The best and most accurate diagnostic method here is angiography. Researchers seek to find economical and high-accuracy methods for this purpose. The disease-related features and different data mining techniques are described to increase the accuracy of the diagnosis through one dataset of essential and useful features. Data are collected from 303 suspected cardiovascular patients in Shahid Rajaee Hospital, Tehran. Among the samples, 87 are healthy, and 216 are sick. The features are selected through their optimal subsets of performance, speed of diagnosis, and precision in the first step to determine the severity of coronary artery disease (CAD). This feature selection can predict and promote a learning model. Then the optimal machine learning models are applied to analyze and predict CAD. The accuracy of 99.67% is found in this diagnosis, indicating the highest obtained accuracy in this field. The left anterior descending (LAD), the left circumflex (LCX), and the right coronary artery (RCA) features are diagnosed with high accuracy by using those models. It seems these three features define the CAD and are dependent on angiography. If they are eliminated for the pre-diagnosis situation, the accuracy of CAD will be between 83% to 86% for the new reduced subset of features proposed concerning legible performance reduction.

Keywords—component; Coronary artery disease; Data mining; Feature selection; Cross-Industry Standard Process for Data Mining methodology; Deep Learning.

I. INTRODUCTION

Data mining is the process of discovering hidden knowledge in a dataset. This discipline is the process of extracting patterns from a large dataset by combining methods of statistics and artificial intelligence and database management. Data mining is widely applied in many fields like marketing, fraud detection, science discovery [1]. Every year, 17 million peoples die due to cardiovascular disease [2]. The most common heart disease among all is Coronary Artery Disease (CAD), due to the narrowing of the coronary arteries. Coronary arteries include three vessels, left anterior descending (LAD), left circumflex (LCX), and right coronary artery (RCA), which feed the heart. When stenosis in either LAD, LCX, or RCA arteries is more than 50%, the patient is diagnosed as having CAD [2]. Angiography is the most common means to diagnose CAD. Due to the

complications in individual patients and the high cost of this diagnosis, researchers seek alternative methods [3].

The Decision Tree (DT), the Naïve Bayes (NB), and the Deep Learning (DL) models are applied for heart disease diagnosis [4]. With due respect to the available finding in this context, the authors here attempt to describe the finding by [5-8] to modify the results, consist of applying the Sequential Minimal Optimization (SMO) algorithm in diagnosing CAD, the yield accuracy was 92.09% [5]. The CAD is assessed by the Bagging algorithm according to laboratory characteristics and echocardiography, on LAD, LCX and RCA with 79.54%, 61.46%, and 68.96% accuracies, respectively [6]; the new features of CAD are tested through the SMO algorithm where the yield accuracy was 94.06% [7] and yielding an accuracy of 96.40% on CAD through a new algorithm [8].

So, it is vital to consider new models for CAD prediction. An attempt is meant here to design a model with high-performance for the diagnosis of imbalanced CAD, in addition to contributing to clinical decisions, reducing errors, as well as saving time and expenses, through high accuracy.

The rest of this paper is organized as follows: the methodology of the study is introduced in Section 2; the experimental plan is described in Section 3, the experimental results are discussed in Section 4, and the article is concluded in Section 5.

II. RESEARCH METHODOLOGY

There exist many methods for implementing data mining projects, among which the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology is the most applied [9]; here, the CRISP-DM is applied as well according to the flowchart of Fig. (1) inspired by [9].

As observed in Fig. 1, the cyclic flowchart of the research methodology of CRISP-DM will be described as follows.

A. Business Understanding

The Impact and risk factors for CAD introduced by [3, 10] are expressed as Hypertension, Diabetes Mellitus, High cholesterol, Obesity, Family history of cardiovascular disease, Dyslipidemia, Use of cocaine and similar stimulants, Cigarette.

CAD is a result of constriction or coronary artery spasm, which is the primary source of blood supply to the heart

muscle [11]. The image vessels causing the CAD [12], is shown in Fig. 2. Exercises like treadmill tests, echocardiograms, nuclear heart scans, and angiography are the ways to diagnose this disease [13], which are usually with errors [14], painful, time-consuming and costly for the patients [15]. So, the development and presentation of specialized systems for the diagnosis of heart disease, especially CAD, are of significant concern [16].

B. Data Understanding

The Cleveland Heart Disease Dataset is usually used for CAD [17]. There exist many symptoms to diagnose heart disease. The patients provide them to diagnose the causes thereof. Here, a series of data from 303 (87 healthy and 216 with CAD) patients of Shahid Rajaei Hospital in Tehran, Tehran cardiovascular center department collected and applied here for their description and modification [7].

C. Data Preparation

The objective here is to increase the quality of the applied data for preparing an appropriate dataset for subsequent phases and modeling. A set of activities should be run on the data. The results of which is then to be transmitted as input to the next stage [9], consisting of (i) Data Selection, (ii) Data Clearing, (iii) Deleting the Unnecessary Fields, (iv) Deleting Duplicate Records, (v) Data Transformation, and (vi) Data Revision.

1) Data Selection:

The dataset is evaluated as to the full features and selected features. The CAD is a target feature (class feature) and exists in all evaluation scenarios.

a) Feature Selection:

The weight functions like weight by Information Gain and weight by Gini Index contained in the RapidMiner tool are applied for feature selection. In Figs. 3-4, the high-weight features are bar charted as the selected features.

• Weights of Information Gain Ratio:

Information Gain measures the reduction in entropy of the data records because of a single split over a given attribute. The entropy before and after the split is computed through. (1):

$$IG = - \sum_{c \in \text{Classes}} P(c) \log(P(c)) \quad (1)$$

Where c is the class value, which can be CAD or Normal, and $P(c)$ is the probability of a record being in class c . Higher values of Information Gain indicate the preference of feature for discrimination of class values [18]. Information Gain Ratio (IGR) is a normalized information gain based on the entropy of values in each class. The output weight features by Information Gain Ratio are bar charted in Fig. 3 for 18 features. The results of IGR show that other features are very negligible, and maybe their removal has no side effect on CAD detection, which in the following sections, this hypothesis will be examined.

• Weight of Gini Index:

Gini Index is a measure indicating how often a randomly chosen element from a set of elements would be incorrectly labeled if it is randomly labeled according to the distribution of labels in the subset. In this setting, the probability of correctly labeling an item is equal to the probability of

choosing the same. Therefore, the Gini Index is computed through (2):

$$GI = 1 - \sum_{c \in \text{classes}} P(c)^2 \quad (2)$$

similar to the Information Gain, the higher values of reduction in the Gini Index indicates that a feature is a better candidate in the classification task [18]. The output of Weight by the Gini Index is shown in Fig. 4 for 21 top features.

The features in Figs. 3 and 4 are selected in this research for CAD detection, and other features from 58 features are removed based on their lower importance. Also, the 18 features of these 21 features without LAD, LCX, and RCA called *selected features* in the following.

2) Data Clearing:

Data clearing is run to identify the missing values. This dataset, unlike the old UCI dataset, has no missing values. Therefore, the precision yield by this dataset is more reliable.

3) Deleting the Unnecessary Fields:

The Remove correlated attributes and Remove useless attributes operations in RapidMiner tools are applied to delete additional fields, which do not find and remove any unnecessary fields.

4) Deleting Duplicate Records:

If there is a patient with duplicate information, the duplicate should be deleted. In this study, no duplicate record is found.

5) Data Transformation:

Sometimes it is needed to transform data, e.g., nominal fields to numerical or vice versa for support vector machines, in which their inputs and outputs can be just numerical. Also, it is possible to convert numerical attributes to nominal or polynomial values, as did in [7]. Still, the Information Gain Ratio and the Gini index of the transformed attributes will be reduced as our experiments show, so we do not use proposed transforms in prior studies. The normalized value of the weighting of the Gini Index and Information Gain Ratio of transformed and non-transformed features are shown in Fig. 5. Never the transformed features (black bars) has more IGR or GI than non-transformed features (white bars). So, it is better to do not to use prior proposed transform approaches.

6) Data Revision:

This phase is for double-checking the earlier phases to find whether any phase should be repeated.

D. Modeling

The following machine learning models are applied in this study:

1) Decision Tree Model:

This model is one of the reliable and commonly applied methods for classification and prediction, due to its transparency, intelligibility, flexibility, and relatively fast structure processing. The prediction derived from this tree is tabulated in a Table and a Tree. In this Tree, every internal node contains a question based on a specific attribute, and there exists a child for each possible answer, where every leaf is labeled with one of the possible classes. The Decision Tree classifies a sample, beginning from the root, by following a path based on the questions posed by the internal nodes and its responses, until it reaches a leaf, finally, which is the class label.

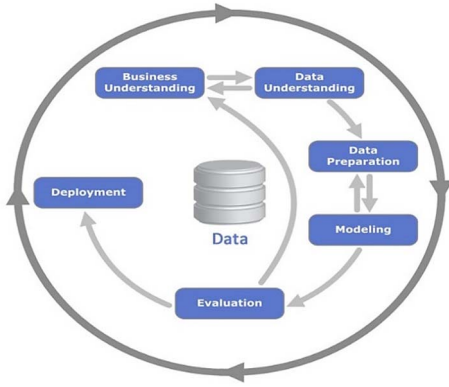


Fig.1. Phases of the CRISP-DM reference model

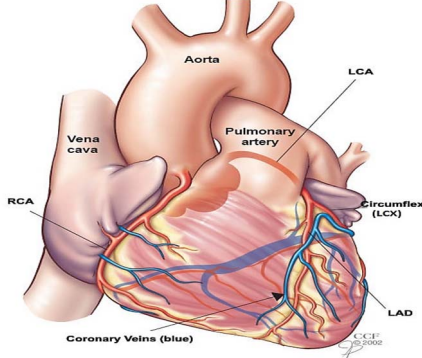


Fig.2. Vessels causing the Coronary Arteries Disease [12]

2) Naive Bayes Model:

Naive Bayes (NB) classifier is a simple probabilistic classifier based on applying Bayes' theorem with strong independence assumptions. The probability of data record X having the class label C_j is calculated through (3). The class label C_j with the most considerable conditional probability value determines the category of the data record.

$$P(C_j|X) = \frac{P(X|C_j) * P(C_j)}{P(X)} \quad (3)$$

3) Deep Learning

Deep learning allows computational models that are composed of multiple processing layers to learn representations of data with multiple levels of abstraction. These methods have dramatically improved state-of-the-art speech recognition, visual object recognition, object detection, and many other domains such as drug discovery and genomics. Deep learning discovers intricate structure in large data sets by using the backpropagation algorithm to indicate how a machine should change its internal parameters that are used to compute the representation in each layer from the representation in the previous layer [25]. The H2O model is one of the effective models of Deep Learning algorithms [26, 27].

E. Evaluation of Performance of Model:

In the cross-validation evaluation model, the data of the train section generate the model, and the data of the test section evaluated the model. In order to evaluate the models, the primary evaluation criteria applied in the field of medicine are of concern. These criteria include [19]:

1) Confusion Matrix

A confusion matrix is a table that allows visualization of the performance of a learning model. In a two classes problem

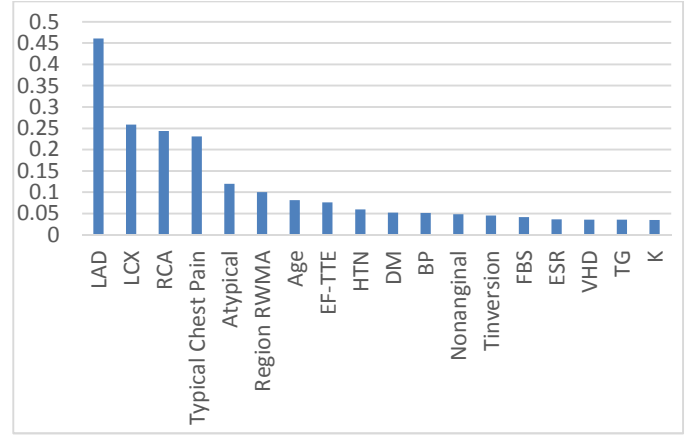


Fig.3. Weights of Information Gain Ratio for features of the dataset

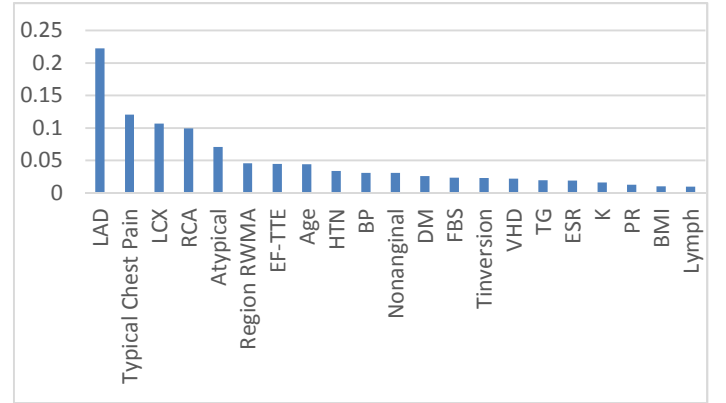


Fig.4. Weight of Gini Index for features of the dataset

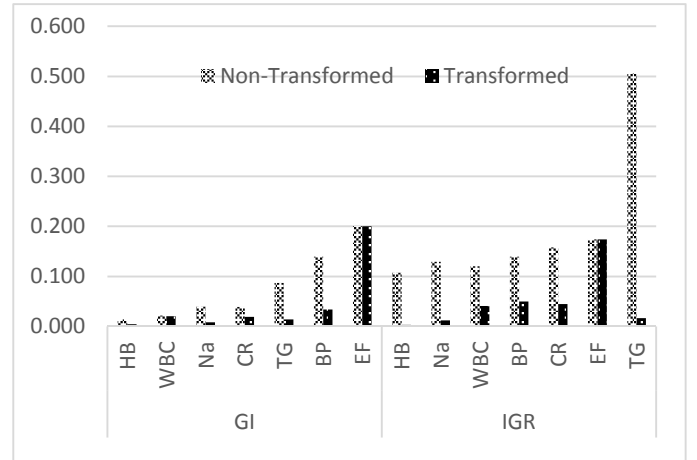


Fig.5. Weight of Gini Index and Information Gain Ratio of transformed features versus non-transformed features

with classes C_1 and C_2 . Table I show that this matrix is of two columns and two rows.

2) Accuracy

The accuracy obtained through. (4):

$$\text{Accuracy} = \frac{(TN+TP)}{(TN+TP+FN+FP)} \quad (4)$$

3) Sensitivity and Specificity:

According to confusion matrix, sensitivity and specificity are explained as follows:

a) Sensitivity (True Positive Rate or Hit Rate or Recall): the probability that a test result will be positive in

the presence of the disease, expressed as a percentage through (5):

$$\text{Sensitivity} = \frac{TP}{(TP+FN)} \quad (5)$$

b) Specificity (True Negative Rate or Selectivity): the probability that a test result will be negative in the absence of the disease, expressed as a percentage through (6):

$$\text{Specificity} = \frac{TN}{(TN+FP)} \quad (6)$$

F. Deployment

At this stage, it is sought to identify the most influential factors in the diagnosis of CAD after issuing the reports and to determine which factors are more efficient in isolating the healthy from the diseased.

III. EXPERIMENTAL PLAN

The proposed plan of the data mining process for CAD detection is shown in Fig. 6, in which the contribution of this approach is using a data selection or dimension reduction heuristic based on IGR and GI, which is highlighted with ** in the figure, to improve the runtime performance metrics of pre-diagnosis CAD. The elements of the proposed approach are described in the Methodology section in detail. It is a part of this research contribution, as mentioned in the introduction section.

IV. EXPERIMENTAL RESULTS

The RapidMiner Version 7.6.3 tool is applied to run different data mining models. The evaluations are run through the Decision Tree (DT), the Naive Bayes (NB), and the Deep Learning (DL) algorithms. Also, k-fold cross-validation is used which evaluate every experiment multiple time to be sure the results are not biased. The selected K for this operator was 10, which cause split the dataset into ten parts. There were 303 records of patients [17], and different data mining models are applied to improve the diagnosis of CAD disease on the UCI dataset [17].

A. The results of different models' performance in the same conditions

According to Table II, the results of the models are described below in various scenarios as following:

- Scenario 1: Results for all features (58 features) based on the proposed data mining model of Fig. 6
- Scenario 2: Results of Alizadeh Sani for all features [7] as a baseline
- Scenario 3: Result for selected features of Alizadeh Sani's with LAD, LCX and RCA features (34 features) based on the proposed data mining model of Fig. 6
- Scenario 4: Results of the approach of Alizadeh Sani for selected features with LAD, LCX and RCA features [7] as a baseline

In this research, we achieved high accuracy, sensitivity, and specificity rates for diagnosis of CAD based on the above scenarios and Table II results, which demonstrate the results of scenarios 1 and 3 are better than scenarios 2 and 4 as a baseline, respectively.

Fig. 7 shows a bar chart for the accuracy column of Table II. There are two impressive results in Fig. 7.

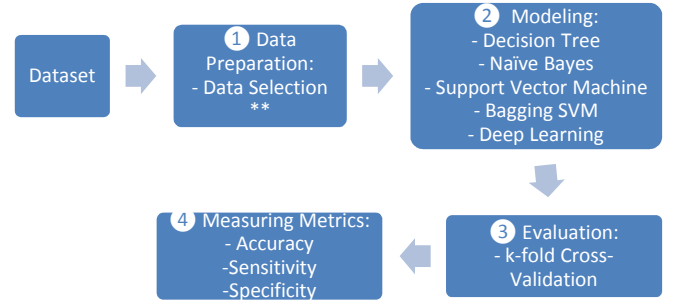


Fig.6. Proposed Data Mining Model for CAD Detection

TABLE I. CONFUSION MATRIX

	Actual class C ₁	Actual class C ₂
Predicted class C ₁	True Positive (TP)	False Positive (FP)
Predicted class C ₂	False Negative (FN)	True Negative (TN)

TABLE II. THE RESULTS OF DIFFERENT MODELS' PERFORMANCE IN THE SAME CONDITIONS AND K=10

Features	Models used	Accuracy	Sensitivity	Specificity
Scenario 1	Decision Tree	99.67%	100%	98.85%
	Naïve Bayes	81.82%	76.85%	94.25%
	Deep Learning	95.42%	99.07%	86.21%
Scenario 2	Neural Network	87.11%	91.67%	75.86%
	Naïve Bayes	63.31%	50%	96.55%
	SVM	90.09%	91.67%	86.21%
	Bagging SVM	90.10%	91.67%	86.21%
Scenario 3	Decision Tree	99.67%	100%	98.85%
	Naïve Bayes	84.77%	79.17%	98.85%
	Deep Learning	98.01%	99.07%	95.40%
Scenario 4	Neural Network	88.11%	91.20%	80.46%
	Naïve Bayes	75.51%	67.59%	95.40%
	SVM	94.08%	96.30%	88.51%
	Bagging SVM	93.40%	95.83%	87.36%

- Scenarios 1 and 3 have better results in both scenarios of all features and selected features against baselines scenarios 2 and 4, respectively.
- The results of selected features can be equal or more than the results of all features. For example, the results of the decision tree are the same in both scenarios. The results of the deep learning model are better and equal for accuracy and sensitivity, respectively, but less for specificity in selected features scenario versus all features scenario, based on Table II and Fig. 7. Finally, the results of Naïve Bayes are better for all metrics in selected features scenarios versus all features scenarios.

The highest accuracy is obtained through the DT, which is 99.67%. This result is the highest accuracy ever achieved in this field. Similar but a lower percentage of 52.33%, 94.08%, 96.40%, 90%, 70% and 75% are obtained by [4, 7, 8, 20-22] which are shown in Fig. 8.

As observed in Table II, the accuracy of the DT model is greater than the other ones obtained through the NB and the DL models. The DT model sensitivity is 100%, indicating that the CAD can correctly be identified by it.

In addition to the high accuracy obtained by the DT model, other results are also found, which they are also of great importance, including:

- The three features, LAD, LCX, and RCA in the dataset, are the main features in diagnosing CAD from NORMAL. By applying the Weight by Information Gain and Weight by Gini Index functions, it is revealed that these three features have the most significant impact on distinguishing CAD from NORMAL, Figs. (3 and 4).
- The DT of scenarios 1 and 3 of Table II is shown in Fig. 9, where if LAD=Stenotic, then it can be said that the person has CAD, otherwise, if LAD=Normal and RCA=Stenotic, then it can be said that the person has CAD and if LAD=RCA=Normal but LCX=Stenotic, then it can be said that the person has CAD. If LAD=RCA=LCX=Normal, then the person is healthy.

The results of scenario 1 for different machine learning models are shown in Fig. 10, which demonstrate using all features, especially LAD, LCX, and RCA can predict CAD very well using all kind of machine learners.

So, in the next section, this research focuses on improving CAD detection based on other features without considering three features LAD, LCX, and RCA.

B. Evaluation of Result for selected features without LAD, LCX and RCA features (18 features)

To evaluating the model, one new scenario is considered as follows:

- Scenario 5: Result for selected features (18 features) of the dataset for simple machine learning models as state-of-the-art

Fig. 11 shows the result of different machine learning models. The deep learning model has the best result. The best result of the state-of-the-art for this dataset is for SVM (SMO) without three features LAD, LCX, and RCA with 93.39% accuracy [7], but when we implement that approach, the results were less than 90 percent even many efforts against the parameters of the evaluation was used, which was irregular.

V. CONCLUSION AND FUTURE WORK

It was found that three features, LAD, LCX, and RCA, have a direct relationship with the diagnosis of heart disease according to the experimental results. In other words, they are the definition of heart disease, and with considering them, there is no need for another feature. Therefore, it is better to recognize the heart disease without considering these three features, because these three features require medical testing. In the early diagnosis of heart disease, it must get these results without medical testing. It is suggested that in the future, based on other features, three features of the LAD, LCX, and RCA can be predicted so that they maybe can be used to increase the accuracy of the diagnosis of heart disease.

So, in the next section of this research, we focus on improving the performance of Coronary Artery Disease (CAD) detection based on prior features and check the effect of dimension reduction in CAD detection. The proposed approach for dimension reduction based on considering Information Gain Ratio and Gini Index shows that it is

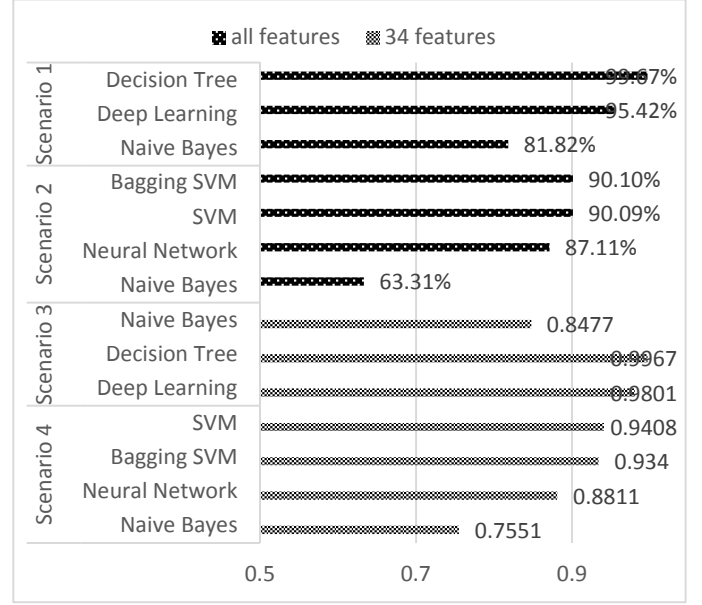


Fig.7. Bar chart of the accuracy of Table II

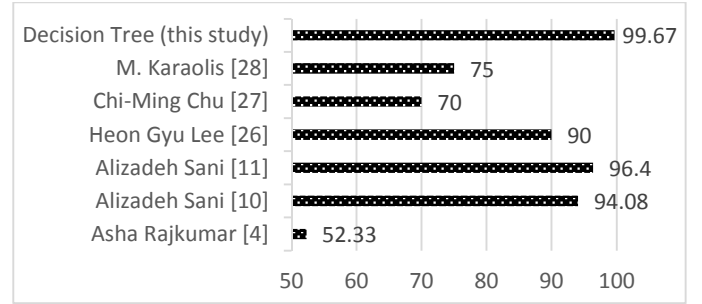


Fig.8. Comparison of the most accuracy obtained in the state-of-the-art

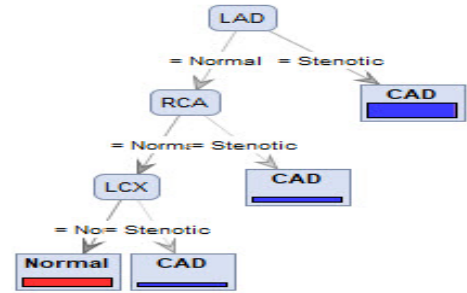


Fig.9. A decision tree with LAD, LCX, RCA, and other features

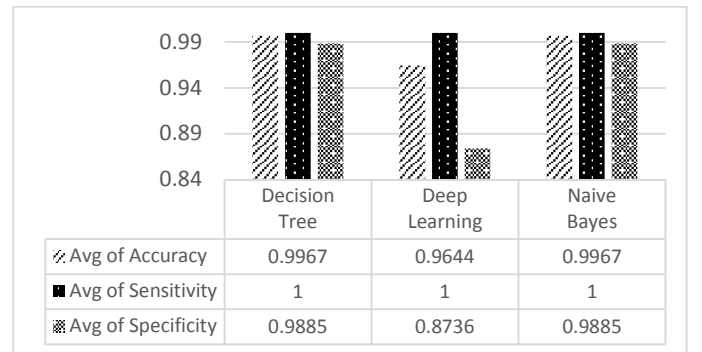


Fig.10. Results of LAD, LCX, and RCA features with different algorithms

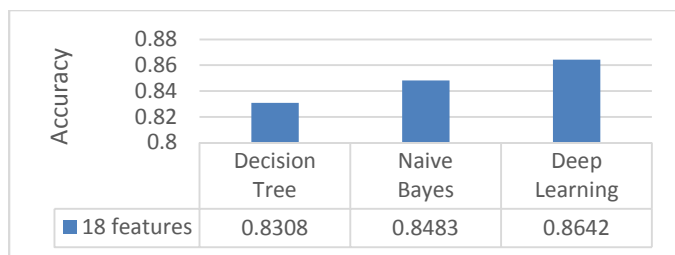


Fig.11. Result for selected features without LAD, LCX and RCA features (18 features)

possible to reduce the 55 features to 18 features with negligible performance reduction. It could be useful for cost-benefit analysis or early CAD detection based on less medical experiments and do more test in very suspicious cases.

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